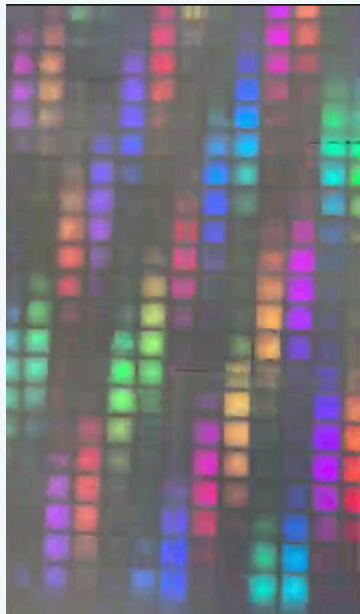


Element G

Initial Design Planning:

Our first idea for the lights was to use the Arduino and individually addressable lights. We did a lot of research on this and even used the FastLED library to create code for our programs. We had different programs for basic light colors and fun sequences. However, over time, this plan started to make less sense for our goals. We were becoming increasingly concerned about our budget and even about completing our project on a realistic timeline. This meant that we needed to switch up our approach. Rather than coding our own lights, we began research on a good pack of pre-coded LEDs that would serve the same purpose and make it a little easier to deliver a functional and budget-friendly product. We then put the lights into a gridded box and began experimenting with different codes.

Photos of initial design / prototype:



```
#include <FastLED.h>
#define LED_PIN    11
#define NUM_LEDS   345
CRGB leds[NUM_LEDS];
void setup() {
  FastLED.addLeds<WS2812, LED_PIN, GRB>(leds, NUM_LEDS);
}
void loop() {
  for(int dot = 0; dot < NUM_LEDS; dot++) {
    leds[dot] = CRGB::Red;
    FastLED.show();
    // clear this led for the next time around the loop
    delay(1);
    leds[dot-1] = CRGB::OrangeRed;
    FastLED.show();
    // clear this led for the next time around the loop
    delay(1);
    leds[dot-2] = CRGB::Yellow;
    FastLED.show();
    // clear this led for the next time around the loop
    delay(1);
    leds[dot-3] = CRGB::Green;
    FastLED.show();
    // clear this led for the next time around the loop
    delay(1);
    leds[dot-4] = CRGB::Blue;
    FastLED.show();
    // clear this led for the next time around the loop
    delay(1);
    leds[dot-5] = CRGB::Indigo;
    FastLED.show();
    // clear this led for the next time around the loop
    delay(350);
    leds[dot-5] = CRGB::Black;
    FastLED.show();
    // clear this led for the next time around the loop
    delay(1);
  }
}
```

Iterated Design Planning:

When we were searching for LED lights to buy in bulk, we actually found that there were none that fit our budget. The only lights that were affordable had apps

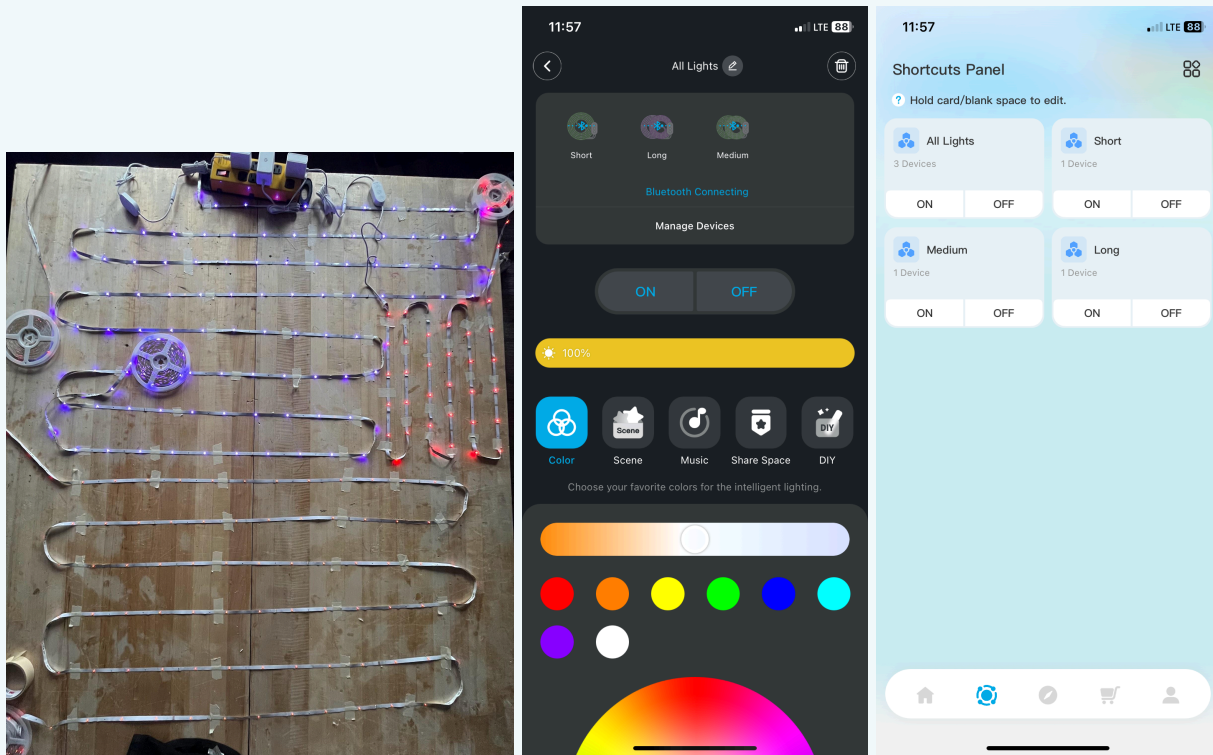
that were already connected to them. We decided to buy these since they were the most efficient option. When we received them, we had many choices of how we could program the lights and change them to match our needs. After trying multiple different configurations and setups within the app, we finally reached consensus. It took a lot of experimenting to figure out how to make the layout easy to understand and use for someone who isn't necessarily coming from an engineering background.

Technical Details & Summarization of Final Design:

We purchased 3 bluetooth RGB LED lights from Govee (100ft, 130ft, and 32.8ft) and worked to create a system on the provided app that would be easy to use. Our group experienced difficulties in terms of how to cross code over multiple devices. We had tried putting all the functions on one of our devices and then played around with signing into other devices or sharing the code across accounts. This was too complicated though, so we decided to secure an iPad that we could give to them permanently and do all the code on there instead.

After experimenting with the Govee app, we were able to figure out what function layout on the app made the most sense and was easiest to follow. We then used a table to make a smaller layout of how the lights will be set up on the ceiling of the classroom. We split up the class into zones: the front, back and sensory room, and have corresponding lights to each section. These lights are fairly simple and do not require much prior knowledge in order to operate.

Photos of Constructed Design:



Video of layout (video):

 IMG_8294.mov

Video of pattern design (prototype) :

 IMG_3732.MOV

Testable Attributes:

When we deliver this product to the teachers at Whitemarsh Elementary, we will be able to understand how functional our design is by simply asking and taking comments into consideration for future ideas. We will also be able to manipulate the colors and designs of the LED lights, thereby allowing us to better understand how entertained the children might be. We can also observe the students' interactions with the lights and take note of how excited they are by them. This would allow us to get a sense of not only the aesthetics, but also the effectiveness of the lights. Perhaps one of the most important ways to evaluate our project is to take a look at affordability and ask important questions like, "Did we stay on budget?" or "How much did we deviate by and for what reasons?" Affordability is both easy to test and

an extremely important criteria for our design. It is also fairly objective, meaning that we should be able to reach a consensus on it without losing too much time. Safety is another concern that we had, and this can easily be tested by seeing how close the kids would be able to get to the lights and making sure that they are not able to reach or tear any down. Overall, these were the main testable attributes of our design that we felt were most crucial to evaluating our project and the ones that were easiest to complete right after installation.

Non-testable attributes:

Something non-testable is the idea of adaptability. In order to test adaptability, we would have to monitor usage and purpose of usage for several weeks on end, which is not a realistic use of time. We won't be in the room at all times that the lights are used, so we will not be able to test how well the lights are truly meeting the needs of the class or adapting to their plans. We have also received feedback from our teacher Mr. Muscarella that this is a fair thing to not be able to test, as we have time constraints and unfortunately, will not be able to complete meaningful tests on this concept. Durability is also difficult to test without long periods of examination, so preliminary testing will not allow for evaluations of durability. However, we are able to say that the lights are definitely dust-proof and water-proof (IP65), as communicated by experts on the product we purchased. Although we cannot validate this assumption ourselves, we are confident in what experts on the LED lights we are using have stated. We are able to numerically score most of our criteria and important attributes, but the ones that require some more time consideration and evaluation are not likely to be testable right after installation. Adaptability and durability are two examples of those.